

HOMEWORK ASSIGNMENT 2

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Instructions:

- Kindly recall that homework assignments for this course are **not turned in**, and **not graded**.
 - However, problems from homework sets will form the basis of problems that appear on the weekly quizzes; as such, it is in your best interest to complete the homework sets on your own
 - Solutions to this assignment will be released at 8:00am on Thursday, September 3 in time for the corresponding quiz on **Wednesday September 9** (note that Monday September 7 is a holiday!)
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1. **Commute Times.** Vanessa wants to determine the average commute time (time taken from place of residence to place of work/school) across all California residents. To collect her data, she goes to the Cal Poly registrar's office, obtains the emails of all currently-enrolled Cal Poly students, and sends a survey to their email addresses asking what their commute times are. 15 Cal Poly students respond to her survey; these 15 students have an average commute time of 7.23 minutes.

- (a) Describe an observational unit in the context of this dataset.
- (b) Define the population of interest.
- (c) Define the parameter of interest. Be sure to use proper mathematical notation!
- (d) What is the sampling frame?
- (e) Did Vanessa obtain a simple random sample? Why or why not?
- (f) Do you think Vanessa's sampling method is biased or unbiased? If you think it is biased, do you think it has a tendency to underestimate or overestimate the true value of the parameter of interest? **Hint:** recall that residents of large, sprawling cities like San Francisco and Los Angeles are also California residents.
- (g) How generalizable do you think Vanessa's findings will be? Why?

2. **Going Gaga for Lady Gaga.** What is the average length (in minutes) of a Lady Gaga song? To answer this question, Morenike went online to repository that maintains a list of all Lady Gaga songs, and assigned a unique number to each Lady Gaga song. She then used a random number generator to generate 20 numbers, and looked up the 20 Lady Gaga songs corresponding to these numbers. These 20 Lady Gaga songs had an average runtime of 3.56 minutes.

- (a) Describe an observational unit in the context of this dataset.
- (b) Define the population of interest.
- (c) What is the sampling frame?

- (d) Did Morenike obtain a simple random sample? Why or why not?
- (e) How generalizable do you think Morenike findings will be? Why?
- (f) Suppose Morenike finds out that the repository she was using actually doesn't contain *all* Lady Gaga songs, and some tracks were not listed. What type of bias does this introduce into Morenike's sample? Make sure you've read all of Lecture Activity 3!
3. **Restaurant Ratings.** Two friends, Agnieszka and Bianca are having a conversation about restaurant ratings. In particular, they are looking at the online reviews for a local restaurant. Agnieszka argues that the average rating listed online is probably lower than the "true" rating, given that people tend to be more inclined to write reviews when they have a particularly negative experience. Bianca argues that, though that may be true, the restaurant has thousands of ratings and therefore the average rating listed online is probably pretty accurate.
- (a) Rephrase Agnieszka's argument in the language of sampling discussed in Lecture Activity 3. In particular, make note of:
- The parameter of interest
 - An observational unit in the context of the problem
 - The population
 - The sampling frame
 - The sampling method
 - Whether Agnieszka is claiming their sampling method to be biased or unbiased, and if it is biased whether she's claiming the method has a tendency to over- or underestimate.
- (b) Is Bianca's argument correct, or is she making a mistake? If she's making a mistake, what mistake is she making? (**Hint:** according to what we saw in Lecture Activity 3, there is a correct answer here!)
4. **Possible Probabilities.** For each of the following values, determine whether it is a valid probability value or not. For those that are not valid probability values, explain why they are not valid probability values.
- (a) 0.00
- (b) 0.8
- (c) 1.25
- (d) -0.43
- (e) 0.2

5. **Coin Conundrum.** Marissa has four coins, but she's concerned that they may not be fair (i.e. that they may land "heads" more or less often than "tails"). To that end, she enlists the help of her friends and tosses each coin 3000 times and records the relative proportion of heads at each stage. Shown below are the relative frequency diagrams associated with each coin (labeled "A" through "D")

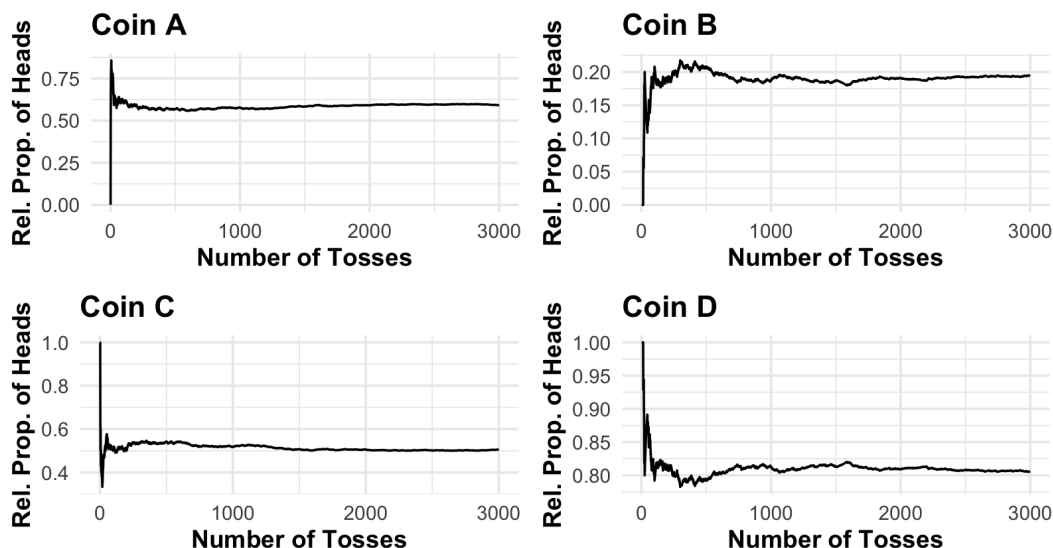


Figure 1: Four relative frequency diagrams.

- (a) What is the probability that Coin A lands "heads?" How can you tell? Is Coin A fair?
- (b) What is the probability that Coin B lands "heads?" How can you tell? Is Coin A fair?
- (c) What is the probability that Coin C lands "heads?" How can you tell? Is Coin A fair?
- (d) What is the probability that Coin D lands "heads?" How can you tell? Is Coin A fair?
6. **Killer Cereal.** *Killer Cereal*, a brand of breakfast cereal, claims that there is a 1 in 4 chance that a box contains a prize. That is; they claim the probability that a box contains a prize is $\frac{1}{4} = 0.25 = 25\%$. Which of the following are valid interpretations of this probability? Select all that apply.
- ☐ Among the next four boxes of **Killer Cereal** you buy, exactly 1 will contain a prize.
 - ☐ In a very large sample of **Killer Cereal** boxes, we would expect around $\frac{1}{4}$ of them to contain a prize.
 - ☐ In a sample of 80 **Killer Cereal** boxes, around 20 would contain a prize
 - ☐ One in every four boxes of **Killer Cereal** will, with absolute certainty, contain a prize
 - ☐ One box of **Killer Cereal** is likely to contain a quarter of a prize